School of Future Tech

Case Study Report

on

ICU Resource Monitoring System

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1. Introduction to the Case Study

In modern hospitals, Intensive Care Units (ICUs) play a critical role in saving lives. ICU patients require constant monitoring and access to specialized resources such as ventilators, hospital beds, and trained medical staff. However, during emergencies or peak medical situations, hospitals often struggle to track the availability and allocation of these resources efficiently.

Manual tracking methods can lead to delays, errors, and mismanagement of resources. For example, a patient may need a ventilator immediately, but due to poor tracking systems, hospital staff may not know whether one is available or already in use.

With the advancement of healthcare technology, hospitals are increasingly adopting digital systems to manage resources efficiently. Real-time monitoring systems help medical staff respond quickly during critical situations.

The **ICU Resource Monitoring System** is designed to simulate a digital system that tracks ICU resources in real time. The system helps hospital staff manage ICU beds, monitor equipment availability, assign patients to resources, and generate reports on resource utilization. This system improves hospital efficiency and helps ensure that critical medical resources are used optimally.

2. Problem Statement / Case Background (Abstract)

Hospitals often face challenges in managing ICU resources effectively, especially during high patient inflow situations such as pandemics, accidents, or medical emergencies. Critical resources such as ICU beds, ventilators, and healthcare staff must be allocated quickly and efficiently to patients in need.

Traditional systems that rely on manual records or basic spreadsheets can lead to delays in resource allocation and poor visibility of available resources. This can negatively affect patient care and hospital efficiency.

This case study focuses on designing and implementing a simulated ICU Resource Monitoring System that allows hospital administrators to monitor resource availability, assign patients to ICU resources, track equipment usage, and generate utilization reports. The system helps ensure better decision-making and efficient use of ICU facilities.

3. Problem Statement / Case Study Design

The ICU Resource Monitoring System is designed with a simple and structured architecture to simulate real-world hospital resource management.

Main Components of the System

The system includes the following modules:

1. ICU Bed Management

This module tracks the availability and occupancy status of ICU beds. Each bed can be marked as available or occupied depending on whether a patient is assigned to it.

2. Equipment Availability Tracking

This module keeps track of critical medical equipment such as ventilators and monitors. It allows administrators to see which equipment is currently in use and which is available.

3. Patient Assignment

When a patient requires ICU care, the system assigns the patient to an available ICU bed and necessary equipment.

4. Usage Reports

The system generates reports showing how ICU resources are being utilized. These reports help hospital administrators understand resource usage patterns and plan improvements.

4. Methods & Algorithms Technology Applied

To build the ICU Resource Monitoring System, several programming concepts and simple algorithms are used.

Programming Language

The system can be implemented using **C++**, which supports object-oriented programming and efficient data management.

Data Structures Used

The following data structures are used in the system:

Arrays / Vectors
Used to store lists of ICU beds, ventilators, and patients.

Structures or Classes
Used to represent hospital resources such as:

- Bed
- Equipment
- Patient

Each object stores relevant information such as availability status and assigned patient.

Algorithms Used

1. Resource Allocation Algorithm

When a patient arrives, the system searches for the first available ICU bed and assigns the patient to it.

Steps:

1. Check list of ICU beds
2. Find the first available bed
3. Assign patient to that bed
4. Mark the bed as occupied

2. Equipment Availability Check

The system checks whether required equipment (like ventilators) is available before assigning it to a patient.

Steps:

1. Check equipment list

2. Find available equipment
3. Assign it to patient
4. Update equipment status

3. Usage Report Generation

The system calculates resource utilization.

Example formula:

Resource Utilization (%)
= (Used Resources / Total Resources) × 100

This helps administrators understand ICU workload.

5. Implementation Details and Snapshots

System Implementation

The system is implemented as a **console-based application** using C++.

The application performs the following functions:

- Add ICU beds
- Track equipment availability
- Assign patients to ICU resources
- Display current ICU status
- Generate usage reports

Example Workflow

Step 1
The system starts and loads ICU resource data.

Step 2
Hospital staff adds patient information.

Step 3
System checks available ICU beds.

Step 4
The patient is assigned to a bed and required equipment.

Step 5
System updates resource status.

Step 6
Admin can view usage reports.

Snapshots-

```
===== ICU Resource Monitoring System =====
1. Admit Patient
2. Discharge Patient
3. View ICU Resources
4. View Patients
5. Usage Report
6. Exit
Enter choice:
```

```
===== ICU Resource Monitoring System =====
1. Admit Patient
2. Discharge Patient
3. View ICU Resources
4. View Patients
5. Usage Report
6. Exit
Enter choice: 1

Enter Patient ID: 123
Enter Patient Name: Rohit
Assign Ventilator? (y/n): y

Patient assigned to Bed 1
```

```
===== ICU Resource Monitoring System =====
1. Admit Patient
2. Discharge Patient
3. View ICU Resources
4. View Patients
5. Usage Report
6. Exit
Enter choice: 3

===== ICU Resource Status =====
Total Beds: 10
Available Beds: 9
Total Ventilators: 5
Available Ventilators: 4
```

```
===== ICU Resource Monitoring System ====
1. Admit Patient
2. Discharge Patient
3. View ICU Resources
4. View Patients
5. Usage Report
6. Exit
Enter choice: 5

===== ICU Usage Report =====
Beds Used: 1/10
Ventilators Used: 1/5
Total Patients: 1
```

6. Results and Conclusion

The ICU Resource Monitoring System successfully simulates the management of ICU resources within a hospital environment.

The system provides an efficient way to monitor and allocate ICU resources such as beds and ventilators. It reduces manual effort and improves visibility of available resources.

Through automated tracking and reporting, hospital administrators can make better decisions regarding resource allocation and patient care.

Although this project is a simplified simulation, the same concept can be expanded into real hospital systems with features such as real-time databases, web interfaces, and integration with hospital management systems.

Overall, the system demonstrates how digital solutions can improve healthcare resource management and enhance operational efficiency in hospitals.

7. References

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